

# Errata to “Unbounded field operators in categorical extensions of conformal nets”

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1. In Sec. 2.7, Def. 2.42, there is a typo:  $\tilde{I}_2$  should be clockwise (rather than anticlockwise) to  $\tilde{I}_1$ . Also, the second paragraph can be more precise. The revised version of Def. 2.42 (with only one paragraph) is as follows.

Let  $\mathcal{E}_{\text{par}}^w$  be a categorical partial extension. Given  $\mathcal{H}_{i_1}, \dots, \mathcal{H}_{i_n} \in \mathcal{F}$ ,  $\tilde{I} \in \tilde{\mathcal{J}}$ , we define  $\mathcal{H}_{i_n, \dots, i_1}(\tilde{I})$  to be the set of all

$$(\mathfrak{a}_n, \dots, \mathfrak{a}_1) \in \mathfrak{H}_{i_n}(\tilde{I}_n) \times \dots \times \mathfrak{H}_{i_1}(\tilde{I}_1)$$

such that  $\tilde{I}_t \subset \tilde{I}$  and  $\mathfrak{a}_t \in \mathfrak{H}_{i_t}(\tilde{I}_t)$  for each  $1 \leq t \leq n$ , and  $\tilde{I}_s$  is clockwise to  $\tilde{I}_t$  whenever  $1 \leq s < t \leq n$ .

2. In the last paragraph of the proof of Thm. 3.39, the formula  $\dim_{\mathcal{C}_V}(U) = \dim_{\text{Rep}^0(A)}(U)$  is not correct, and needs to be deleted. In the subsequent sentence, the formula

$$D(\mathcal{C}_V)/D(\text{Rep}^0(A)) = \dim_{\text{Rep}^0(A)}(U)$$

citing [KO02] Thm. 4.5 is also citing incorrectly. The correct formula in that paper is

$$D(\mathcal{C}_V)/D(\text{Rep}^0(A)) = \dim_{\mathcal{C}_V}(U)$$

These two mistakes indeed cancel each other, which again implies  $D(\mathcal{C}_V)/D(\mathcal{C}_U) = \dim_{\mathcal{C}_V}(U)$  and hence finishing the proof of Thm. 3.39.

3. In the proof of Lem. 3.47, the last sentence “Thus they also hold for  $V_\Lambda \otimes V_L$  and hence for  $V_\Lambda$  by theorems 3.38 and 3.39” is a little misleading. The more appropriate one should be “Thus they also hold for  $V_\Lambda \otimes V_L$  (by theorems 3.38 and 3.39) and hence for  $V_\Lambda$ ”.